TSM 099

# Homework 1

Number structure, powers, radicals, and evaluation

**Topics 1-8** | Estimated time: about 1 hour

**Complete all problems and show your work on separate paper.** Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

Work without a calculator before checking. Give exact, simplified answers.

## Arithmetic foundations

**1.** Use divisibility rules to decide which of 2, 3, 5, 9, and 10 divide 450 and 234. Explain without long division.

**2.** Write the prime factorization of 180 and use it to reduce 126/180.

**3.** Evaluate or state undefined. Explain each zero-denominator case.

$\frac{-42}{-6}$, $\frac{0}{-9}$, $\frac{-9}{0}$, $\frac{0}{0}$

**4.** Find an integer pair for each sum/product: (a) -1 and -20; (b) -11 and 24; (c) 2 and -15.

**5.** Simplify.

$\frac{-84}{126}$, $\frac{-72}{-120}$

## Radicals and signs

**6.** Simplify exactly.

$\sqrt{72}$

**7.** Simplify exactly.

$\sqrt{\frac{49}{81}}$

**8.** Simplify exactly.

$\sqrt{200}$

**9.** Simplify exactly.

$\sqrt{50} + \sqrt{8}$

## Real and complex contexts

**10.** For each expression, state whether it has a real value. Then simplify over the complex numbers using i where needed.

$$\sqrt{-49}, \qquad -\sqrt{49}, \qquad \sqrt{(-7)^2}$$

**11.** Simplify for every real x, including negative inputs.

$$\sqrt{16x^2}$$

## Order, substitution, and equivalence

**12.** Evaluate.

$$-3^2 + 2(-4+1)^2$$

**13.** Evaluate.

$$\frac{18 - 2(3+1)}{5-3}$$

**14.** Evaluate when x = -2 and y = 3.

$$2x^2 - 3xy + y^2$$

**15.** Evaluate both values when a = 1, b = -3, and c = -2. Simplify the radical.

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

**16.** Evaluate both sides at x = -2, 0, and 3. Then decide, with algebraic justification, whether the equation is an identity.

$$(x-2)(x+2) = x^2 - 4$$

**17.** A student claims √(a + b) = √a + √b for nonnegative a and b. Test a = 9, b = 16; explain the conclusion.